

REDWOOD FALLS, MN, USA

WMO: 726556

Lat: 44.548N

Lon: 95.080W

Elev: 1021

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14992

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-8.7	-19.9	1.9	-12.4	-15.4	2.5	-7.9	28.0	14.2	25.6	16.1	9.7	290	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	91.2	74.2	88.3	72.7	85.6	70.8	77.7	87.4	75.5	84.2	73.6	82.0	14.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	135.5	83.5	72.7	125.7	81.4	70.8	117.8	79.4	41.9	87.2	39.8	84.4	37.9	81.9	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.8	22.9	20.0	DB	-19.4	97.1	6.0	2.5	-23.7	98.9	-27.2	100.3	-30.6	101.7	-35.0	103.5	(4)
(5)			WB	-19.6	81.1	6.0	2.3	-23.9	82.8	-27.4	84.1	-30.8	85.4	-35.2	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	14.6	18.7	31.5	46.0	58.8	69.1	73.1	70.3	63.0	49.0	33.8	20.1		
	DBStd	22.95	13.48	13.03	12.91	10.26	8.84	6.65	5.63	5.45	8.37	9.62	11.35	12.15		
	HDD50	4341	1098	877	583	191	23	0	0	0	6	138	497	928		
	HDD65	7727	1563	1297	1038	572	234	33	6	15	135	504	937	1393		
	CDD50	2811	0	0	11	71	295	574	717	629	397	106	11	0		
	CDD65	723	0	0	0	3	41	158	259	179	76	7	0	0		
	CDH74	6971	0	0	4	66	528	1512	2494	1493	764	107	3	0		
	CDH80	2381	0	0	0	17	184	529	921	457	245	28	0	0		
	WSAvg	10.0	10.7	10.9	10.7	11.1	10.7	9.4	8.1	7.5	9.1	10.4	10.6	10.4		
	PrecAvg	26.7	0.7	0.7	1.5	2.6	3.2	4.0	3.9	3.2	2.4	2.0	1.4	0.7		
(15) Precipitation	PrecMax	37.6	1.9	1.7	4.3	7.6	9.6	11.5	7.4	8.3	6.8	7.0	5.3	2.3		
	PrecMin	15.0	0.0	0.0	0.3	0.3	0.5	0.9	0.5	0.9	0.3	0.0	0.0	0.0		
	PrecStd	6.3	0.5	0.4	0.9	1.5	1.8	2.2	1.9	2.0	1.6	1.6	1.3	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	54.4	71.0	81.5	91.7	94.9	95.4	92.2	91.4	84.2	69.1	51.6		
		MCWB	39.0	47.3	56.1	60.8	67.3	73.1	78.6	77.3	71.1	63.3	54.7	46.6		
	2%	DB	39.1	45.3	61.8	74.2	84.9	89.8	91.2	88.3	86.3	76.2	61.7	44.2		
		MCWB	34.9	40.8	52.5	56.2	64.3	72.0	76.2	74.6	70.1	60.6	51.8	38.9		
	5%	DB	35.5	39.6	55.0	68.6	80.0	86.0	88.2	85.1	82.3	70.0	55.0	38.7		
		MCWB	33.0	36.2	48.5	53.3	62.5	69.3	74.7	72.6	67.6	57.5	46.8	35.3		
	10%	DB	32.7	36.1	48.7	63.3	74.9	82.3	85.2	81.9	78.4	64.5	49.6	35.4		
		MCWB	30.9	33.7	43.9	50.6	59.6	67.4	72.2	70.3	65.5	54.9	43.4	33.1		
	0.4%	WB	39.4	47.9	59.9	63.0	71.9	77.1	81.7	80.2	75.5	68.0	58.2	46.1		
		MCDB	46.0	53.6	65.4	75.9	84.2	89.1	90.2	89.0	85.6	76.5	66.4	49.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	40.7	53.8	59.0	67.8	74.7	78.5	76.9	72.8	63.1	52.1	38.9		
		MCDB	38.4	45.4	60.3	70.2	79.5	85.2	88.2	85.3	81.8	72.2	59.6	43.1		
	5%	WB	33.2	36.7	49.2	55.4	64.7	72.2	76.5	74.5	70.5	59.3	47.8	35.5		
		MCDB	35.3	39.6	55.0	65.1	75.7	82.4	85.5	82.1	78.8	68.3	54.5	38.2		
	10%	WB	31.0	33.7	43.7	52.4	61.9	69.9	74.4	72.4	67.8	55.7	43.6	33.2		
		MCDB	32.9	35.9	49.0	61.8	71.7	79.6	82.6	79.5	75.1	63.7	49.6	35.3		
	5% DB	MDBR	16.1	16.9	16.6	20.1	21.0	20.0	20.6	20.4	22.4	20.2	17.1	15.0		
		MCDBR	18.3	20.2	26.0	30.5	29.3	24.8	23.3	23.5	27.4	29.1	26.6	20.0		
		MCWBR	15.1	15.7	17.2	16.5	13.8	11.1	11.3	11.6	13.0	15.7	17.3	15.9		
		MCWBR	17.3	19.1	24.9	25.5	24.6	21.4	21.1	21.0	22.6	25.0	24.7	18.5		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.277	0.306	0.362	0.384	0.399	0.406	0.410	0.367	0.320	0.281	0.262		
		taud	2.470	2.425	2.457	2.341	2.342	2.357	2.320	2.286	2.383	2.511	2.541	2.474		
	Ebn at Noon		279	292	297	285	281	276	272	267	271	272	269	267		
		Edh at Noon	22	27	30	37	39	38	39	39	33	25	20	20		
(44) All-Sky Solar Radiation	RadAvg		582	911	1202	1487	1719	1916	2073	1708	1341	877	572	431		
	RadStd		63	106	121	149	130	142	71	136	99	115	51	43		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page